B1.3 Defined conversion mechanism

This section defines the requirements to convert any legal AXI transaction for use on an AXI4-Lite component. *Conversion, protection, and detection* on page B1-127 discusses the advantages and disadvantages of the various approaches that can be used.

B1.3.1 Conversion rules

Conversion requires that the AXI data width is equal to or greater than the AXI4-Lite data width. If this is not the case then the AXI data width must first be converted to the AXI4-Lite data width.

Note

AXI4-Lite does not support EXOKAY responses, so the conversion rules do not consider this response.

The rules for conversion from a full AXI interface are as follows:

- If a transaction has a burst length greater than 1 then the burst is broken into multiple transactions of burst length 1. The number of transactions that are created depends on the burst length of the original transaction.
- When generating the address for subsequent beats of a burst, the conversion of bursts with a length greater than 1 must take into consideration the burst type. An unaligned start address must be incremented and aligned for subsequent beats of an INCR or WRAP burst. For a FIXED burst the same address is used for all beats.
- Where a write burst with length greater than 1 is converted into multiple write transactions, the component responsible for the conversion must combine the responses for all of the generated transactions, to produce a single response for the original burst. Any error response is sticky. That is, an error response received for any of the generated transactions is retained, and the single combined response indicates an error. If both a SLVERR and a DECERR are received then the first response received is the one that is used for the combined response.
- A transaction that is wider than the destination AXI4-Lite interface is broken into multiple transactions of the same width as the AXI4-Lite interface. For transactions with an unaligned start address, the breaking up of the burst occurs on boundaries that are aligned to the width of the AXI4-Lite interface.
- Where a wide transaction is converted to multiple narrower transactions, the component responsible for the conversion must combine the responses to all of the narrower transactions, to produce a single response for the original transaction. Any error response is sticky. If both a SLVERR and a DECERR are received then the first response received is used for the combined response.
- Transactions that are narrower than the AXI4-Lite interface are passed directly and are not converted.
- Write strobes are passed directly, unmodified.
- Write transactions with no strobes are passed directly.

  Note

  The AXI4-Lite protocol does not require these transactions to be suppressed.

- The **AxLOCK** signals are discarded for all transactions. For a sequence of locked transactions any lock guarantee is lost. However, the locked nature of the transaction is lost only at any downstream arbitration. For an exclusive sequence, the AXI signaling requirements mean any exclusive write access must fail.
- The **AxCACHE** signals are discarded. All transactions are treated as Non-modifiable and Non-bufferable.

  Note

  This is acceptable because AXI permits Modifiable accesses to be treated as Non-modifiable, and Bufferable accesses to be treated as Non-bufferable.

- The **AxPROT** signals are passed directly, unmodified.

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- The **WLAST** signal is discarded.
- The **RLAST** signal is not required, and is considered asserted for every transfer on the read data channel.

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